CS 306 – Database Systems
Project Phase 1

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Project Title:
Gym Management System

Description:
This project involves developing a comprehensive database system to manage the daily operations and administrative activities of a growing gym network. Gym facilities handle a massive amount of information every single day, such as keeping track of their various branches, staff members, registered customers, and physical fitness machines. The main goal of this database is to collect all this fragmented information into one centralized and organized platform. By doing this, the gym management can easily see which trainer works at which branch, where specific equipment is currently located, and which members are actively attending group workouts. This centralized tracking not only makes the daily management of the gym much easier and safer, but it also helps in preventing data loss and scheduling conflicts.

The Branch table stores location details, including the branch ID, name, city, and phone number. This helps to manage different gym sites easily. The Employee table keeps staff information, such as SSN, name, phone, job title, and salary, which is needed for personnel tracking. The Equipment table holds machine details, including equipment ID, type, condition, and purchase date, to keep track of the physical items inside the gyms.

The Member table records user data, such as member ID, name, age, and address. The Class table stores workout session information, including class ID, name, capacity, and duration, to organize group activities properly. Finally, the Emergency_Contact table keeps safety phone numbers for the members. Together, these tables create a simple and organized database to manage the daily tasks of the gym.

ER Model

Entities and Attributes:
The database consists of 6 core entities. The details of these entities are listed below:

1. **Branch:** Stands for the physical gym facilities in different locations.
- A single entry stores data for one specific gym site.
- **Attributes:** bid (Primary Key), bname, bcity, bphone.

2. **Employee:** Contains information regarding the gym staff and management.
- A single entry keeps the personal and job details of a distinct worker.
- **Attributes:** ssn (Primary Key), ename, ephone, job_title, salary.

3. **Equipment:** Keeps track of the physical fitness machines.
- A single entry records the status of one piece of training equipment.
- **Attributes:** eqid (Primary Key), type, condition, purchase_date.

4. **Member:** Holds the details of the registered gym clients.
- A single entry defines one unique customer.
- **Attributes:** mid (Primary Key), mname, mage, maddress.

5. **Class:** Represents the organized group training sessions.
- A single entry details a specific group workout event.
- **Attributes:** cid (Primary Key), cname, capacity, duration.

6. **Emergency_Contact:** Stores safety contact details for customers. Since it relies on a member to exist, it is designed as a Weak Entity.
- A single entry is an emergency phone number tied to a specific member.
- **Attributes:** ecname (Partial Key), ecphone, relation.

Relationships:

1. **Works_In Relationship:**
- Type: Many-to-One (M:1).
- Meaning: An employee is assigned to only one branch, while a branch employs multiple people.
- Visual: A single diamond connecting the Employee rectangle to the Branch rectangle.

2. **Contains Relationship:**
- Type: Many-to-One (M:1).
- Meaning: A machine is placed in exactly one branch, while a branch houses many machines.
- Visual: A single diamond connecting the Equipment rectangle to the Branch rectangle.

3. **Instructs Relationship:**
- Type: Many-to-One (M:1).
- Meaning: A fitness session is led by exactly one trainer, while a trainer can lead multiple sessions.
- Visual: A single diamond connecting the Class rectangle to the Employee rectangle.

4. **Enrolled Relationship:**
- Type: Many-to-Many (M:N).
- Meaning: A customer can participate in several classes, and a class accommodates multiple customers.
- Visual: A single diamond linking Member to Class. It also has an enrollment_date attribute attached directly to the diamond.

5. **Has_Contact Relationship:**
- Type: One-to-Many (1:N) Identifying Relationship.
- Meaning: A member can have multiple emergency contacts, but each contact belongs solely to that specific member.
- Visual: A double-lined diamond linking the Member rectangle to the Emergency_Contact double-rectangle.

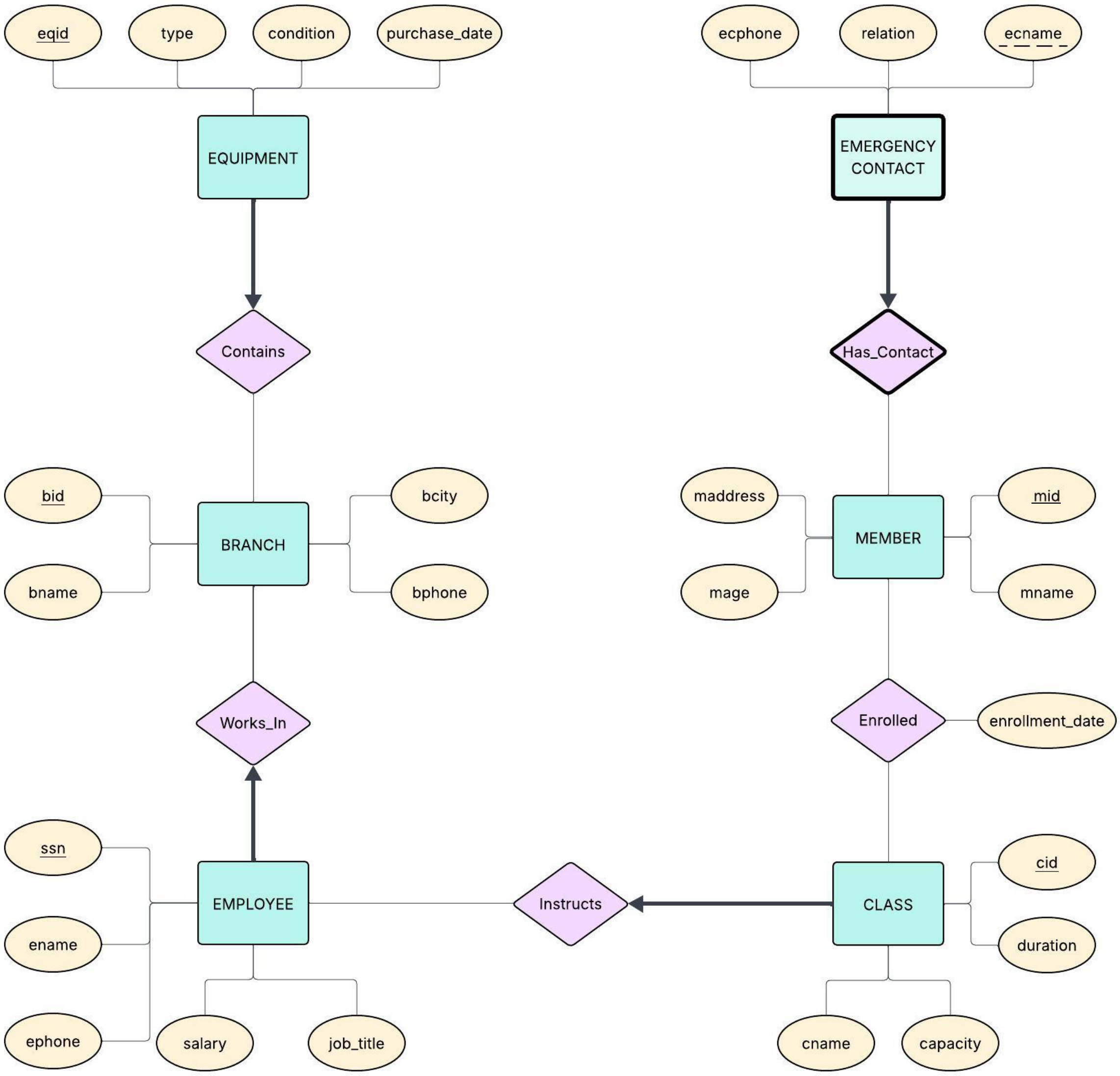
Participation and Key Constraints:

- **Employee Assignment Constraint:** Every employee must be assigned to exactly one branch. (Total participation and key constraint on the Employee side) so *bid* is *Foreign Key* in Employee entity.

- **Equipment Placement Constraint:** Every piece of equipment must be located in exactly one branch. (Total participation and key constraint on the Equipment side) so *bid* is *Foreign Key* in the Equipment entity.

- **Class Instructor Constraint:** Every class must be taught by exactly one instructor. (Total participation and key constraint on the Class side) so *ssn* is *Foreign Key* in the Class entity.

- **Emergency Contact Dependency:** Every emergency contact must be strictly tied to exactly one member. (Total participation and key constraint on the Emergency_Contact side) so *mid* is *Foreign Key* in the Emergency_Contact entity.



- **Appendix (CHAT-GPT)**

Prompt:

- Do these relationship definitions match with our ER diagram? Please answer very shortly. (We added our ER diagram and our project description to the prompt.)

Response:

- Yes, the relationship definitions match your ER diagram.

Briefly:

1. Works_In (Employee → Branch) – M:1 ✔

Each employee works in one branch; a branch has many employees.

2. Contains (Equipment → Branch) – M:1 ✔

Each equipment belongs to one branch; a branch contains many equipment items.

3. Instructs (Class → Employee) – M:1 ✔

Each class is led by one trainer; a trainer can lead multiple classes.

4. Enrolled (Member ↔ Class) – M:N ✔

Members can join multiple classes, and classes can have multiple members.
enrollment_date correctly belongs to the relationship.
5. Has_Contact (Member → Emergency_Contact) – 1:N Identifying ✔
A member can have multiple emergency contacts, and each contact depends on one member.
Conclusion: All relationship definitions are consistent with the ER diagram.